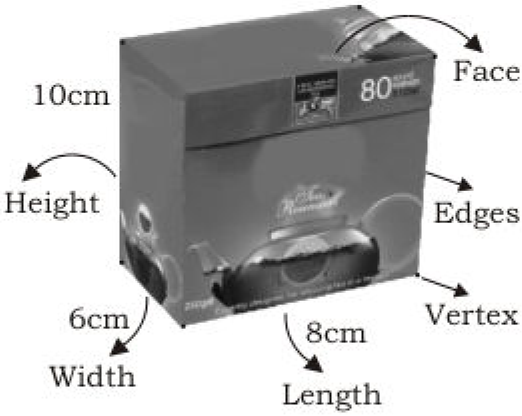
UNIT 12 THREE DIMENSIONAL SOLIDS

VOLUME AND SURFACE AREA

In daily life we come across with solid objects like dice, tea box, tin of ghee, football, funnel etc.

All these objects are three dimensional figures or 3D figures, because they have three dimensions length, width and height.

Let us consider a 3D object tea box as shown here. It has eight vertices, six faces and twelve edges.

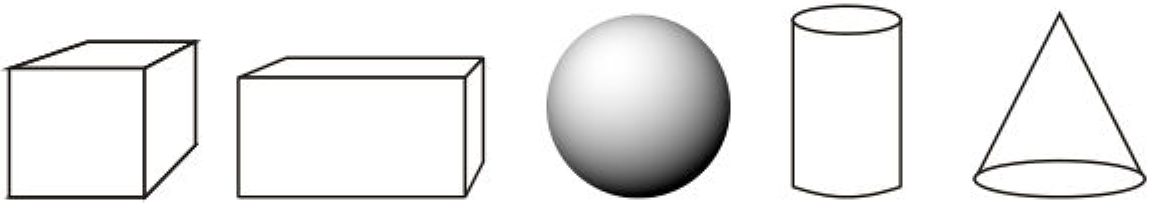


It has length of 8 cm, width of 6 cm and height of 10 cm.

So, it is three dimensional solid figure.

Identify 3D figure (cube, cuboid, sphere, cylinder and cone) with respect to their faces, edges and vertices

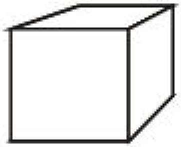
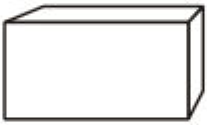
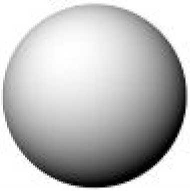
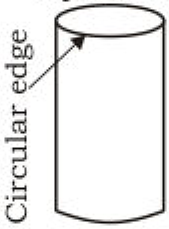
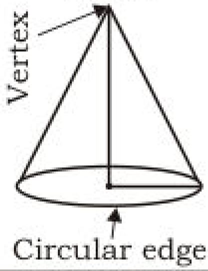
There are many solid figures but we will discuss the properties of five basic 3D figures cube, cuboid, sphere, cylinder and cone. These are shown below:



Cube Cuboid Sphere Cylinder Cone

Teacher's Note *Teacher should give some more examples of solid figures from daily life.*

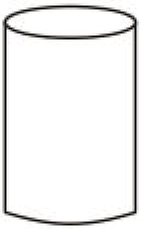
Three Dimensional Figures

3D Figure	Property	No. of Faces	No. of Edges	No. of Vertices
Cube 	It is a solid figure whose length, width and height are equal	6	12	8
Cuboid 	It is a solid figure whose length, width and height are not equal	6	12	8
Sphere 	It is a solid figure with complete round surface.	No	No	No
Cylinder 	It is a solid figure with two opposite circular faces and a curved surface.	2 circular faces	2 Circular edges	No
Cone 	It is a solid figure with one circular face and a curved surface.	1 Circular face	1 Circular edge	1

Teacher's Note *In order to teach 3D-figures, teacher should use available 3D-objects.*

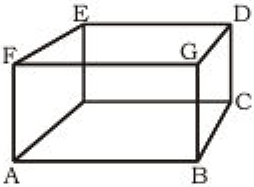
UNIT 12 THREE DIMENSIONAL SOLIDS
(Volume and Surface Area)

Activity **Name the following 3D shapes and fill in the blanks.**

(1) 

Name of figure **Cylinder**

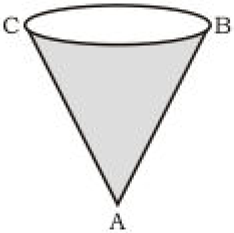
Number of circular faces = _____

(2) 

Name of figure **Cuboid**

$\overline{DE}$ represents _________

Points A, B represent _________

(3) 

Name of figure **Cone**

Point A represents ________

Shaded portion represents _______________

EXERCISE 12.1

1. Identify the objects and write the names of the 3D-figures which represent these objects.

(i) 

(ii) 

(iii) 

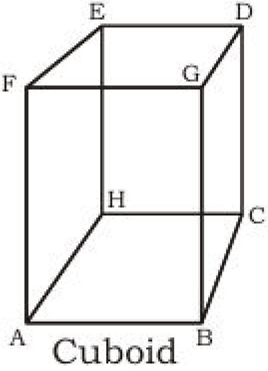
(iv) 

(v) 

(vi) 

2. Fill in the blanks for the following figures.

(i)



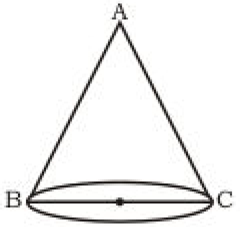
Cuboid

(a) Number of edges = ________

(b) Number of faces = ________

(c) Number of vertices = ________

(ii)

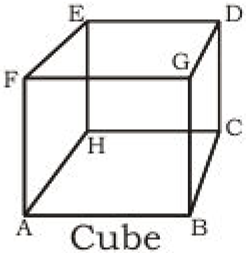


Cone

(a) Point A represents ____________

(b) Number of circular faces ________

(iii)



Cube

(a) $\overline{AB}$ represents ____________

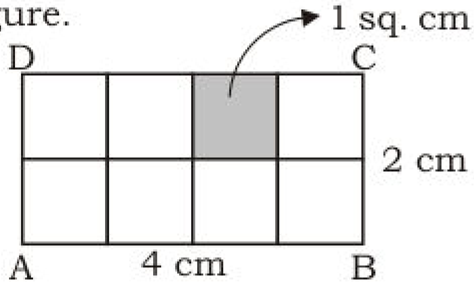
(b) Point A represents ________

(c) Number of faces ________

Define and recognize units of surface area and volume

1. Surface area:

We know that area is the measurement of space occupied by a plane closed figure. Area is found by filling the space by square of one unit length as shown in the figure.



Area of square of one unit length is 1 square unit or 1 sq. unit.

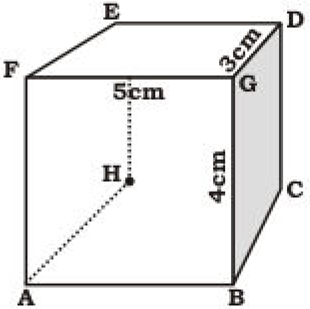
There are 8 squares of area 1 sq. cm.

So, area of rectangle = 8 square cm
= 8 sq. cm

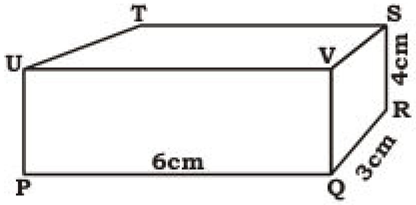
Similarly surface area is the measurement of space occupied by the surface of 3D figure which is also measured in square units.

Note: In measurement of surface area of cube and cuboid, we use square of same unit which is used in length, width and height.

Some examples of units of area are sq. cm, sq. m, sq. km etc.



We will use sq. cm to measure surface area of given cube because its length, width and height are given in centimetres.

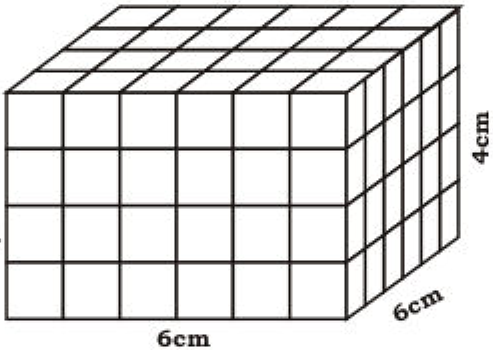


Similarly, we will use sq. cm for given cuboid to measure the surface area.

2. Volume:

We know that volume of solid object is the measurement of the space occupied by a 3D-object. Volume is measured by fitting the cubes of one unit length in the space as shown in the figure.

There are 144 cubes of volume of 1 cubic cm each



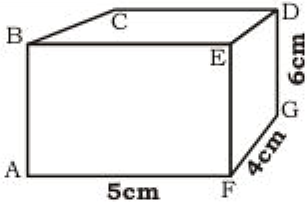
So,
Volume of cuboid
= 144 cubes of volume of 1 cubic cm.
= 144 cubic cm.

Note:

In measurement of volume of cube and cuboid, we use cubes of same unit which is used in length, width and height.

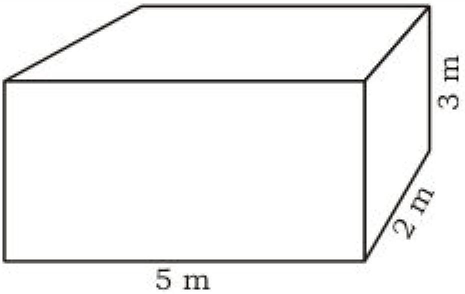
Some units of volume are cubic cm, cubic m, cubic km etc.

For example: In the given cuboid we will use cubic cm to find its volume because its length, width and height are given in centimetres.



Similarly,

The volume of given cuboid will be measured in cubic m.



Activity: Fill in the blanks.

(i) The volume of cuboid of dimensions 5 m, 6 m, 7 m is measured in **cubic m**.

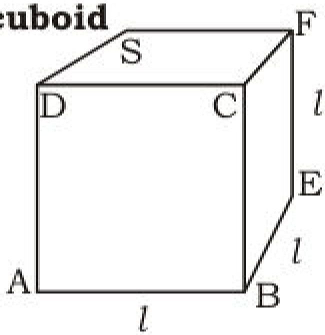
(ii) The volume of cube of each side 6 cm is measured in ________.

(iii) The surface area of cuboid of dimension 6 m, 7 m, 8 m is measured in ________.

(iv) The surface area of cube of length 5 cm is measured in ________.

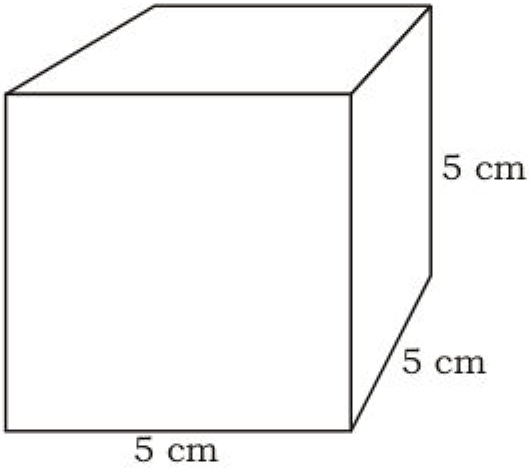
Find surface area and volume of cube and cuboid

1. Surface area of cube



We know that length, breadth and height are equal in cube.
So, each surface is of same area.
Surface area of cube = Area of 6 equal faces
$= 6 \times$ Area of face
$= 6 \times (l \times l)$
$= 6\,l^2$

Example: Find the surface area of cube with each side of 5 cm.

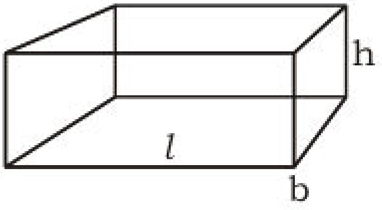


Solution:
Here
Length $= l = 5$ cm
We know that
Surface Area $= 6\,l^2$
$= 6 \times (5)^2$
$= 6 \times 5 \times 5$
$= 150$ sq. cm

Teacher's Note *Teacher should teach the concept of surface area with the help of available solid objects, for example chalk box and geometry box.*

2. Surface area of cuboid.

In order to find total surface area of cuboid, we unfold all the six faces as shown in the figure and find the total area of its six faces.



Surface area of faces of cuboid

= Area of 1 + Area of 2 + Area of 3

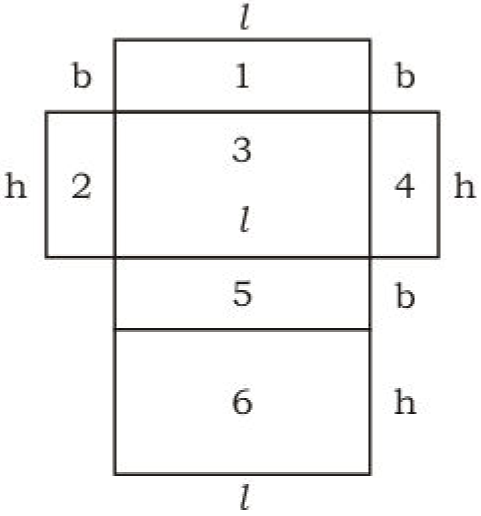
+ Area of 4 + Area of 5 + Area of 6

$= l \times b + b \times h + h \times l + b \times h$

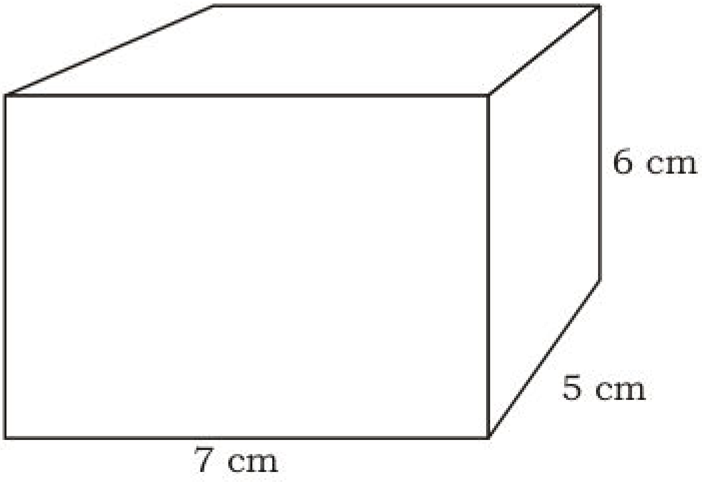
$+ l \times b + h \times l$

$= 2(l \times b) + 2(b \times h) + 2(h \times l)$

$= 2(l \times b + b \times h + h \times l)$



Example: Find the surface area of a cuboid in which
length = 7 cm
breadth = 5 cm
and height = 6 cm.



Solution:

Here
Length = l = 7 cm
Breadth = b = 5 cm
Height = h = 6 cm

Surface area of cuboid $= 2(l \times b + b \times h + h \times l)$

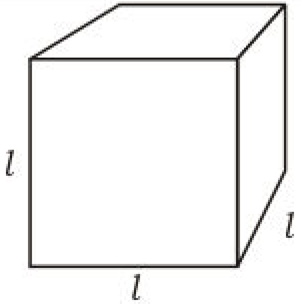
$= 2(7 \times 5 + 5 \times 6 + 6 \times 7)$

$= 2(35 + 30 + 42)$

$= 2(107)$

$= 214$ sq. cm

3. Volume of Cube

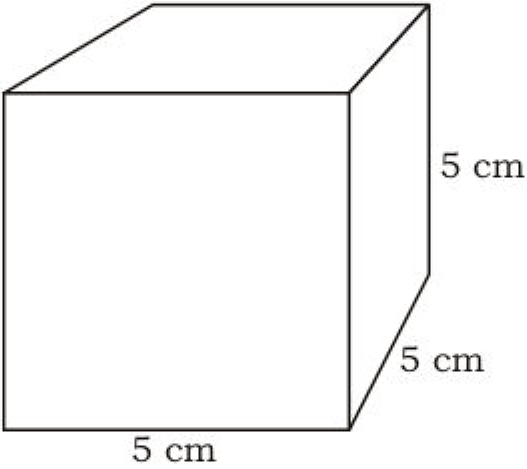


We know that length, breadth and height are equal in any cube.

So, Volume of cube = $l \times l \times l$

$= l^3$

Example: Find volume of a cube whose each side is 5 cm long.



Solution:

Here

Length = l = 5 cm

We know that

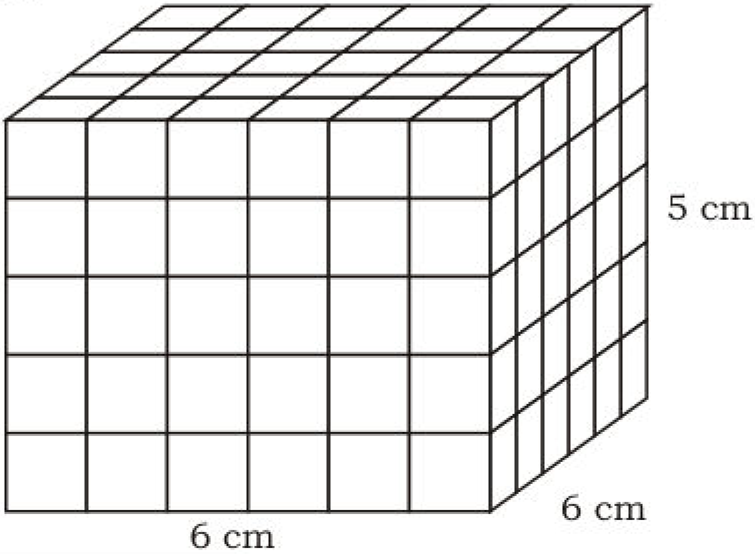
Volume of Cube = l^3

= 5 × 5 × 5

= 125 cubic cm

4. Volume of Cuboid

In order to find formula for volume of cuboid, let us consider a cuboid as shown in the figure.



Here

volume of cuboid

= Area of base × height

= (6 × 6) × 5 cubic cm

= 180 cubic cm

So, $\boxed{\text{Volume of cuboid} = l \times b \times h}$

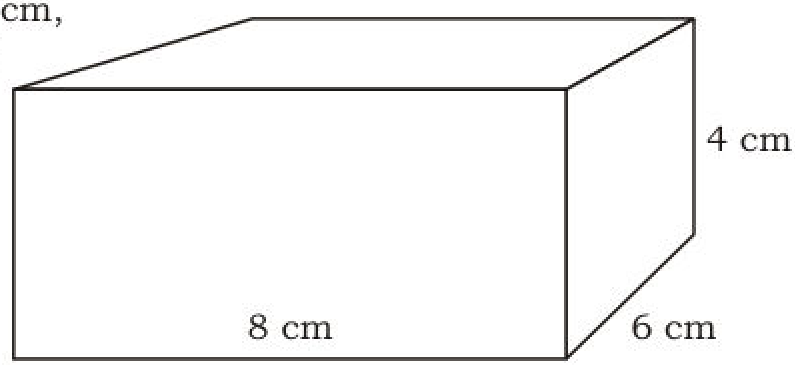
Where

l = Length

b = Breadth

and h = Height

Example: Find the volume of a cuboid in which length = 8 cm, breadth = 6 cm and height = 4 cm.



Solution:

Here

$l = 8$ cm

b = 6 cm

h = 4 cm

We know that

volume of cuboid = $l \times b \times h$

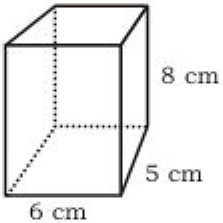
$= 8 \times 6 \times 4$

= 192 cubic cm

EXERCISE 12.2

1. Write the units of surface area and volume of the given figures.

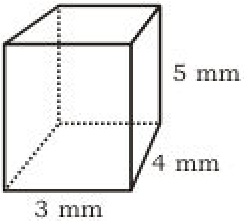
(i)



Unit of surface area = ____________

Unit of volume = ______________

(ii)



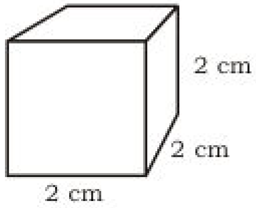
Unit of surface area = ____________

Unit of volume = ______________

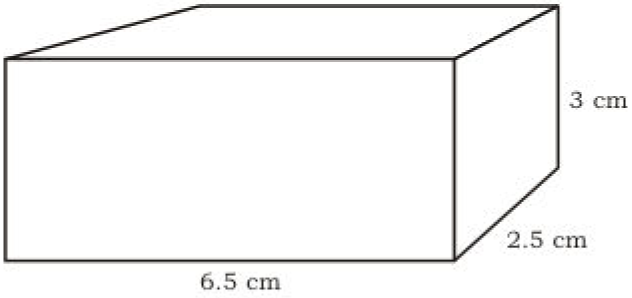
Teacher's Note *Teacher should use solid objects like juice packet and milk packet to teach concept of volume.*

2. Find the surface area and volume of each of the following.

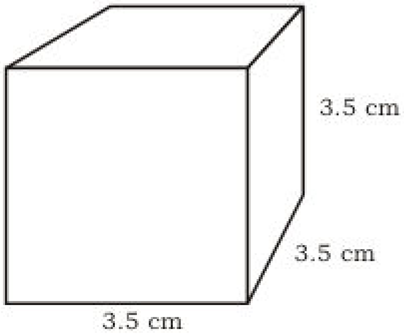
(i)



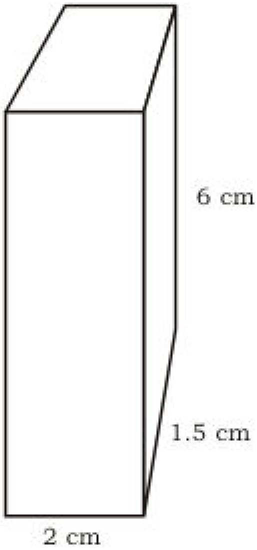
(ii)



(iii)



(iv)



3. Find the volume of the following cubes whose each side is:

(i) 5 cm (ii) 6.2 cm (iii) 2.5 m

4. Find the volume of each of the following cuboids in which:

(i) Length = 9 cm, breadth = 6 cm, height = 4 cm
(ii) Length = 5 cm, breadth = 3.5 cm, height = 4 cm
(iii) Length = 3.5 cm, breadth = 2 cm, height = 2.5 cm

Solve real life problems involving volume and surface area

Example 1: Find the cost of painting a wooden cuboid at the rate of Rs 50 per sq. m if dimensions are 6 m, 5 m and 4 m.

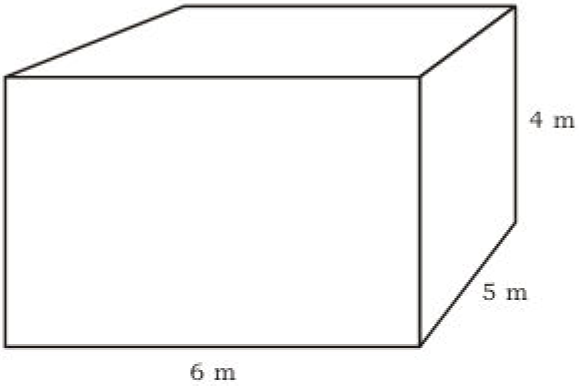
Solution:

Here

$l = 6$ m

b = 5 m

h = 4m



We know that

surface area of cuboid

$= 2\,(l \times b + b \times h + h \times l)$

$= 2\,(6 \times 5 + 5 \times 4 + 4 \times 6)$

$= 2\,(30 + 20 + 24)$

$= 2\,(74)$

= 148 cubic m

Total cost = Surface area x Rate

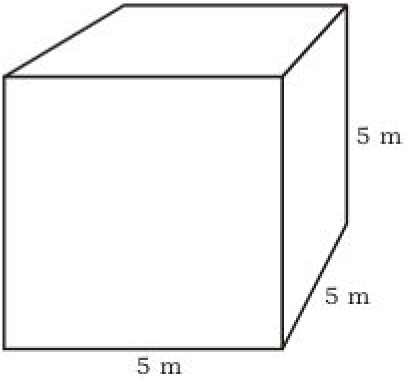
$= 148 \times 50$

= 7400 rupees

Example 2: How much coloured paper is required to cover a cube of side 5 m completely.

Solution:

Here $l = 5$ m



Now

Surface area of cube $= 6\,(l^2)$

$= 6 \times l^2$

$= 6\,(5 \times 5)$

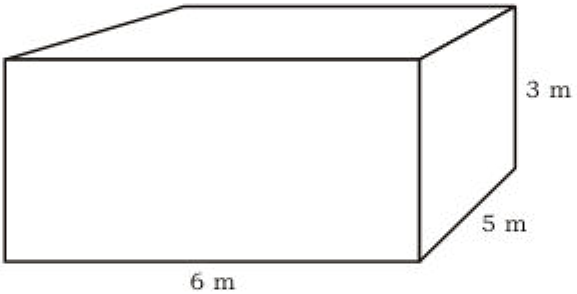
$= 6 \times 25$

= 150 sq. m

Required quantity of paper 150 sq. m

Example 3: Find the capacity of water tank of dimension 6 m, 5 m and 3 m.

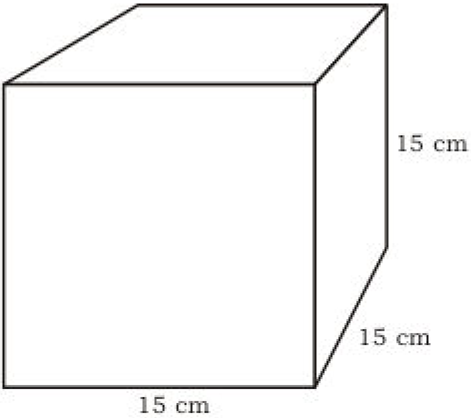
Solution:
Here
$l = 6$ m
b = 5 m
h = 3 m



Now
Volume of cuboid $= l \times b \times h$
$= 6 \times 5 \times 3$
$= 90$ cubic m
So, the required capacity of water tank is 90 cubic m.

Example 4: Find the cost of kerosine oil to fill a cubic tin with each side 15 cm at the rate of Rs 100 per cubic cm.

Solution:
Here $l = 15$ cm



Volume of cubic tin $= l \times l \times l$ or l^3
$= 15 \times 15 \times 15$
$= 5625$ cubic cm
Required cost $= 5625 \times 100$
$= 562500$ rupees

EXERCISE 12.3

1. Find the cost of polishing cupboard with dimension 3 m, 2 m and 1 m at the rate of Rs 500 per sq. m.
2. A tank is 5 m long, 4 m wide and 2.5 m high. Find the capacity of the tank?
3. The dimension of a brick are 25 cm × 10 cm × 4.5 cm. How much space will be occupied by 1000 such bricks?

4. Find the cost of painting a wooden cubic box with an edge of 2.5 m long at the rate of Rs 50 per sq. m.

5. Find the capacity of cubical oil tank with each side 5 m long.

6. How many boxes of dimensions 15 cm × 10 cm × 5 cm are required to fill a box of dimensions 30 cm × 20 cm × 10 cm.

7. If the cost of cake of dimensions 10 cm × 5 cm × 5 cm is Rs 100. What is the cost of the cake with dimensions 20 cm × 15 cm × 15 cm

8. Find the capacity of milk tank of dimensions 3.5 m × 3.4 m × 5.5 m

9. The dimensions of a tea box are $\frac{20}{3}$ cm × $\frac{21}{4}$ cm × $\frac{11}{2}$ cm. How much space will be occupied by 100 such boxes.

10. Find the cost of painting steel cube of each side 5.2 m long at the rate of Rs 15.5 per square metre.

REVIEW EXERCISE 12

1. **Fill in the blanks.**

   (i) Cube has __________ vertices.

   (ii) Cone has __________ circular face.

   (iii) Cuboid has_________ faces.

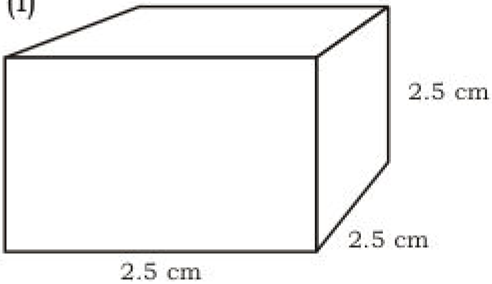
   (iv) Cylinder has _________ circular faces.

   (v) Cube has __________ edges.

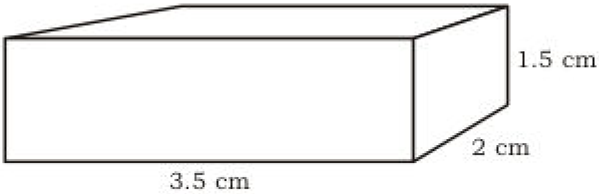
2. Give three examples of 3D figures from daily life.

3. Find the surface area and volume of the following:

(i)



(ii)



4. Find the volume and surface area of a cube whose each side is 1.5 cm long.

5. Find the cost of painting an iron cuboid with dimensions 5 m, 6 m and 7 m at the rate of Rs 100 per sq. m

6. How much milk can be filled in a tank with dimension of 6.5 m 2.7 m and 10 m.

SUMMARY

- Solid objects have three dimensions and are called 3D objects.
- Length, breadth and height of cube are equal.
- Length, breadth and height of cuboid are not equal.
- Cylinder has two circular faces and one curved surface.
- Cone has a circular and a curved surface.
- Sphere has complete round surface.
- Volume of cube = l^3 where l = Length of each side.
- Volume of cuboid = $l \times b \times h$
- Surface area of cuboid = $2(l \times b + b \times h + h \times l)$
- Surface area of cube = $6l^2$